
What is a Dictionary in Python? (Very Simple Explanation)

A **dictionary** is like a **real dictionary** or a **map**.

- It stores data in **key : value** form.
- You search using the **key**, and you get the **value**.

Example in real life:

- Key → "Name", Value → "Srikanta"
 - Key → "Age", Value → 35
 - Key → "City", Value → "Bangalore"
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Python Dictionary Example

```
1 student = {  
2     "name": "John",  
3     "age": 20,  
4     "course": "Python"  
5 }  
6  
7 print(student)
```

Why do we use Dictionary? (Very Simple)

- ✓ To store data in **pairs**
 - ✓ To access values **fast**
 - ✓ To organize data like **records** (student info, product info, bank info)
 - ✓ Keys are **unique**, so no confusion
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How to Access Values (Using key)

```
1 student = {"name": "John", "age": 20}
```

```
2
3 print(student["name"])    # John
4 print(student["age"])    # 20
```

Important Dictionary Functions (Very Simple Examples)

Below are the most commonly used functions and operations.

1 get() — safe way to access values

```
1 student = {"name": "John"}
2
3 print(student.get("name"))    # John
4 print(student.get("age"))    # None (no error)
```

2 keys() — list of all keys

```
1 student = {"name": "John", "age": 20}
2
3 print(student.keys())    # dict_keys(['name', 'age'])
```

3 values() — list of all values

```
1 print(student.values())    # dict_values(['John', 20])
```

4 items() — list of key-value pairs

```
1 print(student.items())
2 # dict_items([('name', 'John'), ('age', 20)])
```

5 update() — add or modify items

```
1 student.update({"age": 21})
2 student.update({"city": "Bangalore"})
3
```

```
4 print(student)
```

6 pop() — remove an item by key

```
1 student = {"name": "John", "age": 20}
2
3 student.pop("age")
4
5 print(student)    # {'name': 'John'}
```

7 popitem() — removes last inserted item

```
1 student = {"name": "John", "age": 20}
2
3 student.popitem()
4
5 print(student)    # {'name': 'John'}
```

8 clear() — remove everything

```
1 student = {"name": "John"}
2
3 student.clear()
4
5 print(student)    # {}
```

9 Add new key-value pair

```
1 student = {"name": "John"}
2
3 student["age"] = 20
4
5 print(student)    # {'name': 'John', 'age': 20}
```

10 Check if a key exists

```
1 student = {"name": "John"}
2
3 print("name" in student)    # True
```

```
4 print("age" in student)    # False
```

Summary (Super Simple)

Operation	Meaning
{}	Create dictionary
d[key]	Access value
get()	Safe access
keys()	All keys
values()	All values
items()	Key+value pairs
update()	Add/modify
pop()	Remove by key
clear()	Empty dictionary